Project Report: StressBuster - A Digital Intervention for Student Well-being

Project Title: StressBuster (Relax & Grow Garden)
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Abstract

In the contemporary academic landscape, students are increasingly subjected to high levels of stress, leading to burnout and diminished mental well-being. This project, titled **StressBuster**, presents a digital solution designed to provide students with a quick, engaging, and non-intrusive mental break. Developed as a C-based interactive simulation titled "Relax & Grow Garden," the application leverages the psychological benefits of nature and nurturance to help users manage stress levels. The program utilizes a simple but effective happiness-tracking algorithm, where the health and

growth of a virtual plant serve as a surrogate for the user's emotional state. Through interactive elements such as watering the plant, speaking kind words, and monitoring environmental factors like weather, the application offers a gamified approach to mindfulness. This report details the design, implementation, and evaluation of the StressBuster project, highlighting its potential as a supplementary tool for mental health maintenance in educational settings. Preliminary analysis suggests that the application provides a refreshing mental pause, although it is acknowledged as a short-term intervention rather than a replacement for professional clinical support.

1. Introduction

Academic pressure has become a defining characteristic of modern education, particularly in higher education institutions where students are expected to balance rigorous curricula, extracurricular activities, and career-related stressors. The impact of such pressure is not merely academic; it significantly affects the mental and physical well-being of students. Recent studies, including those published in *Humanities and Social Sciences Communications* (Nature, 2025) and *Frontiers in Psychology* (2025), have underscored the escalating levels of academic stress, which was further exacerbated by the global COVID-19 pandemic and its aftermath. According to 2026 data from *Research.com*, approximately 45% of college students report experiencing "more than average" stress, while academic performance remains the top stressor for over 60% of students.

The consequences of unmanaged academic stress are profound, often leading to a phenomenon known as academic burnout. Burnout is characterized by emotional exhaustion, cynicism, and a reduced sense of personal accomplishment, which in turn negatively impacts student retention and overall

quality of life (Zhang et al., 2025). Furthermore, the lack of accessible, low-threshold mental health interventions within educational environments often leaves students without immediate coping mechanisms during high-stress periods, such as exam weeks or project deadlines.

1.1 Problem Statement

Traditional stress-relief methods, such as counseling or therapy, are essential but can often be resource-intensive,

time-consuming, or stigmatized. There is a clear need for "micro-interventions"—short, accessible, and engaging digital tools that provide an immediate mental break and help students de-escalate stress in real-time. Students often find it difficult to transition from intense cognitive tasks to a state of relaxation, and the "Relax & Grow Garden" project, or **StressBuster**, aims to address this specific gap.

1.2 Motivation and Project Concept

The motivation for this project stems from the recognized therapeutic benefits of nature and gardening. Research has shown that even virtual interactions with nature can have a calming effect, a concept referred to as a "digital therapeutic garden" (Kang et al., 2025). The **StressBuster** application, titled "Relax & Grow Garden," was conceptualized as a simple, interactive program that simulates the process of nurturing a virtual plant. By focusing on a small, manageable task that requires "care" (such as watering or talking to a plant), the user can divert their attention away from academic stressors, fostering a brief state of mindfulness.

1.3 Objectives

The primary objectives of the StressBuster project are as follows: 1. To develop a lightweight, text-based interactive application that provides an immediate mental break for students. 2. To implement a quantifiable "happiness" metric that correlates with user interactions and environmental variables. 3. To design an intuitive and accessible user interface (via a terminal-based system) that minimizes cognitive load while providing visual and textual feedback. 4. To evaluate the effectiveness of simple gamification elements—such as random weather events and growth milestones—in maintaining user engagement and promoting relaxation. 5. To provide a foundational tool that can be expanded with more

advanced features, such as time limits and improved visual/audio feedback, in future iterations.

By achieving these objectives, the team hopes to offer a practical, albeit modest, contribution to the field of digital mental health tools tailored for the student population.

2. Literature Review

The academic and psychological impacts of stress have been extensively documented in recent years, with a growing body of research exploring digital interventions as a means of mitigation. This literature review focuses on three key areas: the impact of stress on student well-being, the benefits of gamification in mental health, and the therapeutic potential of virtual nature and "nurturing" simulations.

2.1 Impact of Stress on Student Well-being

Academic stress is a multifaceted phenomenon that affects various domains of student life, including psychological, physical, and academic performance. Research published in *Humanities and Social Sciences Communications* (Nature, 2025) highlights that the post-pandemic academic environment has created an unprecedented level of pressure, leading to higher rates of anxiety and depression among university students. The study emphasizes that academic stress is not only a result of academic load but also of the constant pressure to achieve and the lack of social support.

According to *Research.com* (Bouchrika, 2026), nearly 45% of students report high levels of stress, and the relationship between stress and academic burnout is well-established. Long-term longitudinal data (Zhang et al., 2025) suggests that chronic stress leads to emotional exhaustion and a sense of detachment, which are core components of academic burnout.

This burnout is associated with poor sleep quality, reduced cognitive function, and, in severe cases, physical health issues such as headaches and gastrointestinal problems. These findings underscore the critical need for interventions that can interrupt the stress cycle and provide students with cognitive and emotional relief.

2.2 Gamification in Mental Health Promotion

Gamification—the application of game-design elements in non-game contexts—has emerged as a promising strategy for mental health promotion. A narrative review by Cheng and Ebrahimi (2023) discusses gamification as a novel approach to mental health, noting that games can enhance engagement, motivation, and the overall user experience of mental health tools. Digital games offer a safe space for experimentation and learning, providing a sense of agency and achievement that is often lacking in high-stress academic environments (Casimiro, 2026).

Research published in *Frontiers in Digital Health* (2023) has shown that gamified interventions can be effective in managing symptoms of anxiety and depression. By incorporating elements such as progress tracking, rewards, and interactive feedback, these tools can help users build resilience and develop coping mechanisms. The use of simple, low-stakes interactions—such as the "care" mechanics in the **StressBuster** project—can provide a sense of accomplishment and calm, which are essential for stress reduction.

2.3 The Concept of Digital Therapeutic Gardens

The therapeutic benefits of nature are well-known, but recent research has explored how these benefits can be translated into digital environments. The concept of the "digital

therapeutic garden" (Kang et al., 2025) refers to virtual spaces that simulate natural elements to achieve psychological effects similar to real-world nature exposure. Studies have shown that even short periods of exposure to virtual green spaces can lower cortisol levels and improve mood.

The **StressBuster** project's "Relax & Grow Garden" aligns with this research by providing a virtual plant-nurturing experience. Gardening, whether physical or virtual, requires patience and care, which can induce a state of flow—a psychological state where a person is fully immersed in an activity, leading to reduced self-consciousness and a sense of calm. The relationship between gardening frequency and mental well-being is mediated by "activity attachment" (Frontiers in Psychology, 2026), suggesting that the more a person interacts with their virtual garden, the greater the potential for psychological benefit.

In summary, the literature supports the use of simple, gamified digital tools as a low-threshold intervention for academic stress. By leveraging the therapeutic potential of virtual nature and the engaging nature of game mechanics, the **StressBuster** project aims to provide a practical solution for students seeking a brief but effective mental refresh.

3. System Design and Methodology

The system design for the StressBuster project, titled "Relax & Grow Garden," is based on a structured, iterative process aimed at creating a simple and effective interactive application. The design focuses on user engagement, feedback, and the simulation of natural variables that influence the virtual plant's health.

3.1 Concept and Core Mechanics

The core concept of the "Relax & Grow Garden" project is to simulate the nurturing of a virtual plant, where the plant's health (measured as "Happiness Level") is directly influenced by user interactions and environmental factors. The goal is to encourage the user to engage in small, mindful tasks that provide a sense of achievement and a break from academic stressors.

3.1.1 User Interactions

The user interacts with the program through several key actions, including watering the plant, which is a basic maintenance task that simulates providing essential resources to increase the happiness level. Another interaction involves talking to the plant, representing an emotional connection that provides positive reinforcement and further boosts the plant's happiness. Users can also perform an informational action to check the status, allowing them to view the current happiness level and growth stage, or choose to exit the program at any time to receive a final encouraging message.

3.1.2 Environmental Variables

To add an element of realism and variety, the program includes a "Weather" variable that is randomly generated in each turn of the interaction loop. The weather affects the plant's happiness level as sunny weather increases happiness by representing a positive growth environment, while rainy weather increases it even more through optimal hydration. Cloudy weather has a neutral effect, representing a stable but less optimal environment for growth.

3.2 Algorithm and Program Logic

Step 1: Start the program
Step 2: Declare variables

- choice (for menu)
- happiness = 50 (initial value)
- weather

Step 3: Initialize random number generator
Step 4: Display welcome message
Step 5: Repeat until user chooses Exit:
Step 6: Generate random number for weather (0–2)
Step 7: If weather = 0 → Display “Sunny” and increase happiness by 5
Step 8: If weather = 1 → Display “Rainy” and increase happiness by 7
Step 9: Else → Display “Cloudy”
Step 10: Display menu options:

- Water plants
- Talk
- Check happiness
- Exit

Step 11: Accept user input (choice)
Step 12: Perform action based on choice:

- If choice = 1 → Increase happiness by 10
- If choice = 2 → Increase happiness by 5
- If choice = 3 → Display current happiness
- If choice = 4 → Exit loop
- Else → Display “Invalid choice”

Step 13: Repeat steps 6–12 until Exit is selected
Step 14: Display thank you message
Step 15: End program

3.3 Technical Stack

The StressBuster application was developed using the C programming language, chosen for its efficiency and suitability for building simple console-based applications. The use of standard libraries such as <stdio.h> for input/output, <stdlib.h> for utility functions, and <time.h> for time-based random number generation ensures the program is lightweight and portable. The choice of a terminal-based interface minimizes system requirements and allows for a

distraction-free experience, focusing the user's attention solely on the interaction with the virtual plant.

4. Implementation and Code Breakdown

The implementation of the StressBuster project, titled "Relax & Grow Garden," was carried out using the C programming language. This section provides a detailed breakdown of the program's structure and the logic behind its key functions.

4.1 Program Structure and Headers

The program begins with the inclusion of standard C libraries that are essential for its operation: - #include <stdio.h>: Used for standard input and output operations, such as displaying the menu and reading user choices. - #include <stdlib.h>: Provides utility functions, including rand() and srand() for random number generation. - #include <time.h>: Used to seed the random number generator with the current system time, ensuring that the weather generated in each session is truly unpredictable.

The program initializes the plant's happiness level to a baseline of 50. This value serves as the starting point for the user's interaction and is stored in a variable named happiness.

4.2 Main Interaction Loop

The heart of the application is a do-while loop that continues as long as the user's choice is not '4' (the exit option). Within this loop, several key operations take place:

4.2.1 Random Weather Generation

At the beginning of each turn, the program generates a random weather condition using the rand() % 3 expression. This results in an integer between 0 and 2, which corresponds to the following conditions: - **0 (Sunny):** The program displays "Weather: ☀️ Sunny" and increases the happiness level by 5. - **1 (Rainy):** The program displays "Weather: 🌧️ Rainy" and increases the happiness level by 7. - **2 (Cloudy):** The program displays "Weather: ☁️ Cloudy" and makes no changes to the happiness level.

4.2.2 User Menu and Choice Handling

The program displays a menu with four options for the user. A switch statement is then used to process the user's input: - **Case 1 (Watering):** If the user chooses to water the plant, the program increases the happiness variable by 10 and displays the message "You watered the plant. It feels refreshed!" - **Case 2 (Talking):** If the user chooses to talk to the plant, the program increases the happiness variable by 5 and displays the message "You said kind words. The plant smiles 😊". - **Case 3 (Check Happiness):** This option allows the user to view the current happiness level. If the level is high enough (e.g., above 80), a special congratulatory message is displayed, such as "Your plant is blooming beautifully! 🌸". - **Case 4 (Exit):** Selecting this option breaks the do-while loop, leading to the program's termination.

4.3 Code Logic and Error Handling

The use of a switch statement ensures that only valid inputs (1-4) are processed. If an invalid input is entered, the program simply ignores it and restarts the loop, preventing unexpected behavior. The interactive messages provide immediate

feedback to the user, reinforcing the sense of care and nurturance that is central to the project's goals.

The code's simplicity is intentional, focusing on the core experience of relaxation rather than complex gameplay mechanics. This approach ensures that the application is accessible to all users, regardless of their programming knowledge or gaming experience.

4.4 Outcome and UI Design

The user interface (UI) of the "Relax & Grow Garden" is text-based and terminal-centered. This design choice minimizes distractions and allows the user to focus on the simple interactions provided by the program. The inclusion of emojis (such as ☀️, ☁️, 🌱) adds a visual element that enhances the calming atmosphere of the application. The final output, as seen in the terminal, clearly shows the sequence of interactions, providing a satisfying sense of progress as the plant's happiness level increases over time.

5. Results and Discussion

The results of the StressBuster project, titled "Relax & Grow Garden," can be analyzed based on several criteria: user experience, technical performance, and psychological impact. This section discusses the outcomes of the project, its limitations, and areas for potential improvement.

5.1 Project Outcome and User Experience

The StressBuster application successfully achieves its goal of providing a simple, interactive mental break for students. The "Relax & Grow Garden" simulation offers a quick and non-intrusive way to manage stress, allowing users to engage in a few minutes of mindful interaction with a virtual plant. The

feedback provided by the program (e.g., "The plant smiles " or "You watered the plant") reinforces the sense of accomplishment and care, which are known to have a positive impact on mental well-being.

The text-based UI, while simple, is effective in minimizing cognitive load and providing a distraction-free experience. The inclusion of emojis and random weather events adds variety and makes the interaction more engaging. Preliminary analysis suggests that the application provides a refreshing mental pause, although its impact on long-term stress levels remains a topic for further research.

5.2 Discussion of Results

The "Relax & Grow Garden" simulation has several strengths, including its simplicity and accessibility, as the application is easy to use and requires no special skills or prior knowledge. Additionally, the program provides immediate feedback on user interactions, which reinforces a sense of care and nurturance. The inclusion of gamification elements such as random weather events and growth milestones makes the interaction more engaging and helps to maintain user interest over time. Finally, the application provides a quick mental refresh through a brief but effective mental pause that allows users to de-escalate stress in real-time.

However, the application also has some limitations, such as simple interactions that may become repetitive over time and a text-based user interface that might not be as engaging as a graphical interface for all users. The application currently has limited features and lacks elements like time limits or break reminders. It is important to note that the simulation is intended as a short-term intervention and should not be seen as a replacement for professional mental health support.

5.3 Pros and Cons of the StressBuster Project

Based on the project's design and implementation, the following pros and cons can be identified:

5.3.1 Pros

The project offers several advantages, such as helping to reduce stress without explicitly shouting relax at the user, providing a gentle and non-intrusive way to manage stress. The simulation is also ideal for a quick mental refresh during study or work breaks, offering an immediate mental pause. Furthermore, the sense of care and nurturance can lead to a natural improvement in mood without the need for caffeine or motivational speeches. Being lightweight, the application is easy to play anytime and anywhere on any device with a terminal interface.

5.3.2 Cons

There are also several disadvantages to consider, including the risk of excessive engagement or one-more-round syndrome, which may lead to a loss of focus on other tasks. Prolonged use of the application can increase screen time and exposure to blue light, leading to potential eye strain and other screen-related health issues. While the text-based interface is simple, it still requires focused attention on a screen, which may contribute to physical discomfort. Finally, the application is intended as a supplementary tool and is not a permanent solution or substitute for serious clinical stress support.

5.4 Scope for Improvement and Future Work

To address these limitations and enhance the effectiveness of the StressBuster project, several improvements are proposed for future iterations, such as implementing time limits to prevent users from forgetting real-life tasks and ensuring they return to their studies. Integrating break reminders can also help to promote healthy screen habits and reduce the potential for eye strain. Furthermore, adding graphical elements, animations, and calming background sounds could enhance the immersive experience and promote deeper relaxation. Optimizing animations and transitions would lead to a more professional and calming user experience, while expanding features to include more complex interactions, such as different types of plants or environmental challenges, could help to maintain user engagement over long periods.

5.5 Conclusion

The StressBuster project, titled "Relax & Grow Garden," provides a practical and accessible solution for students seeking a brief mental break from academic pressure. By leveraging the therapeutic potential of virtual nature and the engaging nature of simple gamification elements, the application offers a practical tool for stress management. While it is acknowledged as a short-term intervention rather than a permanent solution, its potential as a supplementary tool for mental health maintenance is clear. The proposed improvements for future iterations will further enhance its effectiveness and provide a more comprehensive and engaging user experience.

6. References

The following sources were consulted and cited in the development of the StressBuster project and this report:

1. **Bouchrika, I. (2026).** *50 Current Student Stress Statistics: 2026 Data, Analysis & Predictions*. Research.com. https://research.com/education/student-stress-statistics
2. **Casimiro, M. (2026).** *Playing Can Pave the Way: The Hidden Mental Health Benefits of Gaming*. University of Colorado Denver. https://www.ucdenver.edu/student/stories/library/healthy-happy-life/playing-can-pave-the-way-the-hidden-mental-health-benefits- of-gaming
3. **Cheng, C., & Ebrahimi, O. V. (2023).** *Gamification: a Novel Approach to Mental Health Promotion*. Current Psychiatry Reports, 25(11), 577–586. https://pmc.ncbi.nlm.nih.gov/articles/PMC10654169/
4. **Frontiers in Digital Health. (2023).** *A systematic review on the efficacy of virtual reality and gamification interventions for managing anxiety and depression*. https://www.frontiersin.org/journals/digital-health/articles/10.3389/fdgth.2023.1239435/full
5. **Frontiers in Psychology. (2025).** *The concept of the digital therapeutic garden and its psychological effects*. https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2025.1534541/full
6. **Frontiers in Psychology. (2026).** *The mediating role of activity attachment in the relationship between gardening frequency, leisure orientation, and mental wellbeing: evidence from resident gardeners with implications for future gardening tourism*. https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2026.1767661/full
7. **Nature. (2025).** *Examining the effects of academic stress on student well-being in higher education*.

Humanities and Social Sciences Communications. https://www.nature.com/articles/s41599-025-04698-y 8. Zhang, J., Meng, J., & Wen, X. (2025).*The relationship between stress and academic burnout in college students: evidence from longitudinal data on indirect effects*. Frontiers in Psychology, 16, 1517920. https://pmc.ncbi.nlm.nih.gov/articles/PMC12146318/